

FINAL PROJECT

"The Importance of Sleep for Health"

Deadlines

- **First Submission (Part A):** Friday, July 17, 13:00
- **Second Submission (Part B):** Wednesday, July 22, 13:00
- **Presentations (Part C):** Friday, July 24

General Guidelines

- Groups may consist of **3-4 students**.
 - Your project should not only apply data analytics techniques but also **interpret the results within the context of the problem**.
 - The file name must include the last name(s) of the participant(s).
 - It is essential that you **explain and justify your decisions** throughout the project.
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Context

According to the **World Health Organization**, sleep disorders affect nearly **40% of the world's population**.

Multiple studies have shown that sleep quality has a direct impact on both physical and mental health, influencing the risk of cardiovascular disease, concentration, and overall well-being.

Your team works in the **Data Analytics Department** of a healthcare organization that wants to understand which factors are associated with sleep disorders and, ultimately, predict their occurrence.

The dataset includes variables related to sleep, physical activity, stress, cardiovascular indicators, and lifestyle.

Descriptive Variables

- **Person ID:** Unique identifier
- **Gender:** Gender
- **Age:** Age (18 years or older)
- **Occupation:** Occupation
- **Sleep Duration:** Average daily sleep duration (hours)
- **Quality of Sleep:** Self-reported sleep quality (scale from 1 to 10)
- **Physical Activity Level:** Minutes of physical activity per day
- **Stress Level:** Self-reported stress level (scale from 1 to 10)
- **BMI Category:** Body Mass Index category (Underweight, Normal, Overweight, Obese)
- **Blood Pressure:** Systolic/diastolic blood pressure
- **Heart Rate:** Resting heart rate (beats per minute)
- **Daily Steps:** Number of daily steps

Target Variable

- **Sleep Disorder:** Presence of a sleep disorder (Yes or No)
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Part A – Exploratory Data Analysis (EDA)

In this first part, you will perform an exploratory data analysis focusing on the following aspects:

1. Dataset Overview

- Describe the dataset.
- Identify categorical and continuous variables.

2. Data Visualization

- Use appropriate visualizations based on each variable type.
- Describe key trends and identify potential issues.

3. Data Quality Issues

- Missing values
- Outliers
- Irregular cardinality
- Other relevant issues

4. Relationships Between Variables

- Explore meaningful relationships between descriptive variables and the target variable.
- Identify candidate features for the predictive model.

5. Additional Analysis

- What non-obvious patterns or relationships did you discover?
 - What hypotheses can be generated from the exploratory analysis?
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Part B – Predictive Modeling

In this section, you will build a predictive model to identify individuals who may develop sleep disorders.

Use the variables selected during **Part A**, after completing your data cleaning and exploratory analysis.

1. Data Preparation

- Apply any transformations you consider appropriate.

2. Model Development

- Build a baseline model (Decision Tree).
- Develop different versions or variations of the model.
- If desired, you may experiment with multiple types of predictive models.

3. Model Evaluation

Evaluate your models using metrics such as:

- Accuracy
- Precision
- Recall
- F1-Score
- ROC Curve / AUC
- Confusion Matrix

Justify which evaluation metric you consider the most important and explain why.

4. Profit (Gain) Matrix

- Assign a cost or benefit to each outcome in the confusion matrix.
- Discuss the consequences of false positives and false negatives for both patients and the healthcare system.

5. Model Comparison and Selection

- Which model would you choose?
 - Why is it the best option?
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Part C – Final Presentation

Each group will deliver a **10-minute presentation** explaining the project's findings as if presenting to a healthcare organization interested in using the results to improve people's well-being.

The objective is **not** to explain the technical process or the data transformations performed, but rather to communicate the findings in a **clear, concise, and persuasive** manner, highlighting the key insights and recommendations for decision-making.

Suggested Structure

Problem Statement

- Why is it important to study sleep disorders?
- What is the main research question guiding the project?

Key Findings

- Present the most important patterns, relationships, or high-risk groups identified during the analysis.

Predictions and Results

- How accurately can sleep disorders be predicted using the available data?
- What do these results imply in practice?

Impact and Recommendations

- What actions should the healthcare organization consider based on your findings?
- Which types of patients may require closer monitoring or additional support?

Final Reflections

- Project limitations
- Key lessons learned
- Ideas for future research and analysis